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AI-Based early Disease Risk Prediction System Using Machine Learning

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Abstract: Long term diseases such as diabetes and cardiovascular disease are becoming increasingly common worldwide. The big problem is that most people do not find out they have these conditions until they are already serious. At this point, treatment becomes more difficult, expensive, and sometimes too late. If these diseases can be detected early, people have a much better chance of getting help in time and making healthier choices before their condition get worse. This study presents a system that uses computer-based learning to predict whether a person is at an early-stage risk of developing diabetes or heart disease. The system was trained using two health datasets that included information such as age, blood pressure, cholesterol levels, BMI, glucose levels, and lifestyle habits. Several prediction models were tested and compared to determine which model performed best. Processes such as cleaning the data, extracting the most relevant data and balancing the data set were also carried out in order to improve the reliability of the predictions. The best prediction method for diabetes was the support vector machine with F1-score of 0.9499. In case of the cardiovascular disease risk prediction, Logistic Regression turned out to be the best with an F1-score of 0.9722. Both the diabetes risk prediction models were implemented into a web interface where people can type in their health information and obtain the predicted risk value. This type of smart system can help everyone know better about health and prepare earlier.

Keywords: Machine Learning, Disease Risk Prediction, Diabetes Prediction, Cardiovascular Disease, Healthcare Analytics, Streamlit.

2

1 Introduction

Long term disease like cardiovascular disease and diabetes have become a significant contributor to diseases globally. Early prediction of these diseases may help patients initiate preventative treatments to stop the disease before it is fully developed. The current medical diagnosis techniques require high costs and significant amounts of time especially for patients in region without extensive medical support. The alternative of using machine learning to detect the trends of disease risk, by analyzing the patient's data, may be a more viable option.

The system uses a supervised machine learning approach by training data about the patient's medical background and lifestyle and predicting the risk level of the diseases. Various supervised learning algorithms were trained and tested to establish the best prediction model for each of the chronic illnesses. The system includes preprocessing, feature selection, class balancing and threshold adjustment methods.

1.1 Motivations

The reason this project is being undertaken is because the cases of chronic disease are ever increasing, and there is a need to provide accurate and early diagnosis. While most of the healthcare prediction systems predict the presence or absence of a single disease and remain in a research level, this project tries to create a smart and an approachable system that can serve the need for early health care and preventive health management using Machine Learning.

1.2 Objectives

The major goal of this research is to build an AI based early disease risk prediction system to predict diabetes and cardiovascular disease using ML approach on patient's health record data. Furthermore, this study strives to achieve the best prediction performance and reliability with methods such as preprocessing, feature selection, handling imbalance data using SMOTE, cross-validation, and thresholding. Moreover, this project has a scope to deploy the model using a Streamlit based web application so that it could serve healthcare professionals in real-time.

1.3 Original Contributions

Existing frameworks usually predict only a disease at one time. In this work, prediction of two diseases diabetes and cardiovascular disease is done together in a framework which makes it highly applicable for practical usage. The system was developed completely as a whole end-to-end

3

solution. At first, preprocessing of the data was performed. Features of the health data set which are considered important for the model training were extracted so that the model would stay focused on the main goal. Because of an unequal distribution of samples in the dataset, a sampling technique called SMOTE was applied to balance the data for better learning. Several predictive models were implemented and compared with each other for selecting the best one. Prediction threshold was tuned up to improve performance rather than sticking to the initial value. At last, the solution was deployed in a form of web application.

1.4 Paper Layout

The paper has five sections. Section 2 covers literature survey on available researches on disease prediction system and identifies the research gap. Section 3 talks about proposed system architecture, datasets used, pre-processing and machine learning methods applied in the project. Section 4 explains experimental setup, models' evaluation and prediction and performance analysis of proposed model. Section 5 provides the conclusion and limitations with the future work.

4

2 Literature Survey

Machine learning has been widely applied in healthcare systems for early disease prediction and risk assessment. Almustafa (2020) [1] conducted a comparative study of classification algorithms for heart disease prediction, demonstrating the importance of feature selection and appropriate classifiers for cardiovascular risk assessment. Deberneh and Kim (2021) [2] proposed a machine learning framework for diabetes prediction incorporating ANOVA, Chi-Square, and Recursive Feature Elimination (RFE) for feature selection, emphasizing clinical attribute relevance for improved accuracy. Rajendra and Latifi (2021) [3] investigated diabetes prediction using logistic regression and ensemble learning, finding that ensemble models provide improved predictive capability compared to standalone classifiers.

Poonia et al. (2022) [4] developed intelligent diagnostic models for chronic kidney disease using machine learning with feature selection, demonstrating that effective feature engineering improves disease classification. Ahmad et al. (2022) [5] explored heart disease prediction using a mixed machine learning approach combining numerical and categorical clinical features. Hassan et al. (2022) [6] evaluated multiple machine learning classifiers for coronary heart disease prediction, reinforcing the applicability of data-driven approaches in cardiovascular healthcare. Islam et al. (2022) [7] investigated CKD prediction using several machine learning algorithms, showing that such models can assist healthcare professionals in diagnosing kidney-related disorders.

Recent research has further advanced predictive modeling in healthcare. Uddin et al. (2023) [8] proposed a cardiovascular disease prediction framework using multiple machine learning models and integrated datasets to enhance prediction reliability. Nguyen et al. (2023) [9] examined machine learning approaches for assisting physicians in diabetes diagnosis, highlighting their usefulness as clinical decision-support tools. Alzboon et al. (2025) [10] conducted a comparative analysis of machine learning techniques for diabetes prediction, demonstrating continued advancement in predictive modeling and the importance of selecting suitable algorithms for accurate disease risk assessment.

5

3 Proposed System/Model

This paper presents an AI-Based Early Disease Risk Prediction System for predicting the early risk of diabetes and cardiovascular diseases using machine learning techniques. The system integrates preprocessing, feature selection, imbalance handling, model comparison, and real-time deployment to provide accurate and reliable healthcare risk predictions.

3.1 Dataset Description

Two healthcare datasets were used to develop and evaluate the proposed system. The diabetes dataset contains 6000 patient records with features such as glucose level, insulin, BMI, blood pressure, cholesterol, stress level, and physical activity. The cardiovascular dataset contains 5500 patient records with features including systolic blood pressure, cholesterol, heart rate, smoking status, and diet quality score. Both datasets were pre-processed before model training.

3.2 Schematic Layout of the proposed system/model

The proposed system follows a sequential machine learning workflow. Patient data were collected and preprocessed using scaling and encoding methods. Feature selection was then applied to identify important health parameters. The processed data were balanced using SMOTE and used to train different machine learning models. Finally, the best-performing models were deployed using a Streamlit web application for real-time disease prediction.

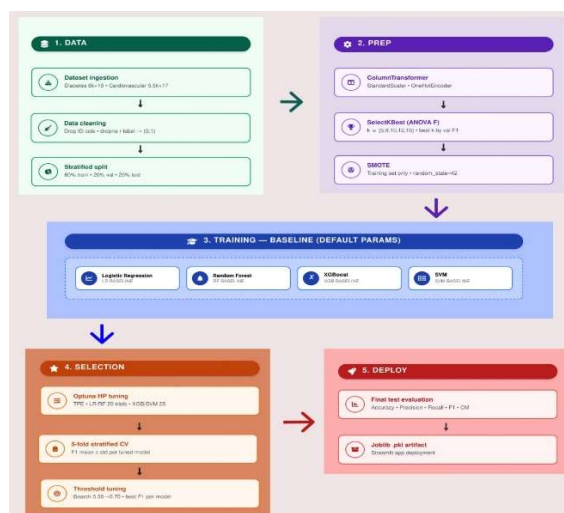


Fig. 1. Schematic Model Diagram of the Proposed Disease Prediction System

6

3.3 Methodologies Used

The proposed system is based on a structured machine learning workflow for health predictions.

- **Step 1: Data Preprocessing:** Missing values, categorical features, and numerical attributes were processed using encoding and normalization techniques to improve data quality and consistency.
- **Step 2: Feature Selection:** SelectKBest with ANOVA F-statistic was applied to identify the most important features for disease prediction.
- **Step 3: Imbalance Handling:** SMOTE (Synthetic Minority Over-sampling Technique) was used to balance minority disease classes and improve prediction reliability.
- **Step 4: Baseline Model Training:** Four classifiers Logistic Regression, Random Forest, XGBoost, and Support Vector Machine (SVM) were trained using default hyperparameters on the balanced training data. The baseline validation performance was recorded using accuracy, precision, recall, and F1-score to establish a reference benchmark before optimization.
- **Step 5: Hyperparameter Optimization (Optuna):** Optuna with Tree-structured Parzen Estimator (TPE) sampling was used to automatically search the hyperparameter space for each model. Logistic Regression and Random Forest were optimized over 20 trials, XGBoost and SVM over 25 trials. The objective function maximized the validation F1-score, avoiding the exhaustive cost of grid search while producing more reliable parameter configurations.
- **Step 6: 5-Fold Stratified Cross-Validation:** Each optimized model was evaluated using 5-fold stratified cross-validation on the training set, reporting F1 mean standard deviation per model. Stratification preserved the original class distribution across all folds, and the averaged F1-score provided a more robust estimate of generalization performance compared to a single train-validation split.
- **Step 7: Threshold tuning and Deployment:** Decision threshold tuning was performed to improve prediction performance, and the final models were deployed using a Streamlit web application for real-time healthcare risk prediction.

4 Experimentation and Model Evaluation

4.1 Experimental Setup

The proposed system was developed using Python 3.10 in a Windows 11 environment with 8 GB RAM. Libraries such as Pandas, NumPy, Scikit-learn, XGBoost, Optuna, imbalanced-learn, Matplotlib, and Streamlit were used in the development. The datasets were divided into 60%, 20%, and 20% training, validation, and testing sets, respectively. SMOTE was applied only to the training data to prevent data leakage. Hyperparameter optimization was performed using Optuna TPE sampler, and 5-fold stratified cross-validation was used to evaluate model generalization.

4.2 Results and Outputs

Table 1. Final Performance Comparison

Disease	Best Model	Accuracy	F1-Score
Diabetes	SVM	94.42%	0.9499
Cardiovascular	Logistic Regression	96.36%	0.9722

The experimental results showed that the SVM achieved the best performance for diabetes prediction, with balanced precision and recall values. Logistic Regression achieved the best overall performance for cardiovascular disease prediction. Threshold tuning improved the prediction balance and reduced incorrect classifications.

4.3 Validation/System Performance Evaluation

Table 2. Diabetes Optimized Model Validation

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9491	0.9612	0.9485	0.9787
SVM	0.9475	0.9611	0.9455	0.9532
XGBoost	0.9458	0.9582	0.9455	0.9518
Random Forest	0.9450	0.9568	0.9455	0.9511

Table 3. Cardiovascular Optimized Model validation

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9718	0.9821	0.9754	0.9787
SVM	0.9600	0.9687	0.9713	0.9700
XGBoost	0.9572	0.9660	0.9699	0.9680
Random Forest	0.9390	0.9487	0.9604	0.9545

SVM was the best model for diabetes prediction. Logistic Regression was the best method for predicting cardiovascular disease. Random Forest achieved the lowest performance in both cases.

4.4 Discussions on Contributions

The experimental findings validate the effectiveness of the proposed AI-based healthcare prediction architecture. The integration of preprocessing, SMOTE balancing, feature selection, hyperparameter optimization, cross-validation, and threshold tuning significantly improved the overall prediction accuracy and reliability compared with conventional machine learning workflows.

The use of separate machine learning pipelines for diabetes and cardiovascular disease allowed the system to adapt more effectively to different healthcare datasets and prediction requirements. In addition, deploying the trained models using a Streamlit-based web application transformed the project from a research-oriented prediction model into a practical real-time healthcare support system.

Overall, the results demonstrate that machine learning can effectively support early disease risk prediction and contribute toward accessible, preventive, and data-driven healthcare systems

5 Conclusion and Future Scope

This project successfully built an AI-based system to predict early risk for both diabetes and cardiovascular disease using machine learning. Four models were trained and compared using standardized preprocessing, class balancing with SMOTE, feature selection, hyperparameter optimization, k-fold cross validation and threshold tuning.

The SVM achieved an accuracy of 94.42% and an F1-score of 0.9499 for diabetes prediction. Logistic Regression achieved 96.36% accuracy and an F1-score of 0.9722 for cardiovascular disease prediction. A user-friendly Streamlit web app was deployed to provide real-time predictions from user entered health datasets.

Current limitations include predictions limited to two diseases, dependency on dataset quality, longer training time with larger datasets, and the system is meant for prediction support only and cannot replace a medical diagnosis.

Future work can extend the system to predict more diseases such as kidney or liver disease, use real-time and larger datasets, add deep learning models such as neural networks, and develop a mobile-friendly version. Adding explainable AI features would also help users understand why a prediction was made.

10

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